

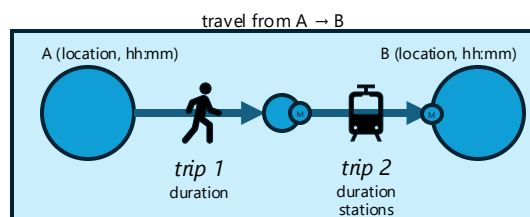
Location intelligence: Mapping spatiotemporal population dynamics in the Mexico City Metropolitan Area for emergency management

Dr María Fernanda Lammoglia Cobo
MPhil in Engineering



How does population distribution change throughout the day in the Metropolitan Area of Mexico City on a weekday and weekend, and how can this information support disaster preparedness?

INEGI 2017 origin-destination (OD) survey

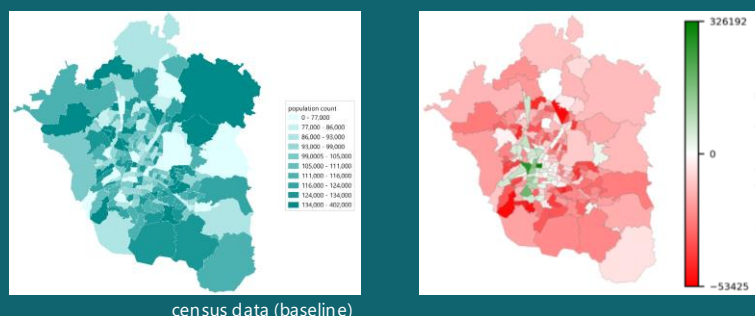


modelling

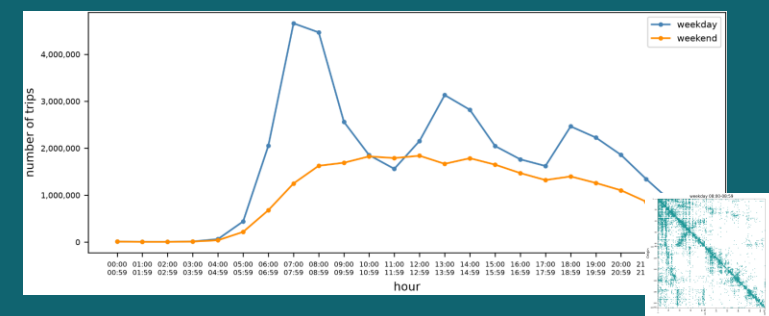
48 OD matrices

origin neighbourhood	destination neighbourhoods									
	1	2	3	4	5	6	7	8	9	10
1	35317	3760	1772	1970	473	1452	3384	780	68	104
2	7064	24002	1693	179	570	165	1379	5735	1060	366
3	6156	3043	16075	812	0	126	726	414	0	176
4	2741	961	1315	8838	1899	2725	266	579	0	106
5	4306	2227	561	2494	9778	2338	1212	619	461	137
6	2707	971	404	2806	566	8650	1523	1517	0	365
7	4394	1195	261	206	144	1144	24090	3337	147	2289
8	340	7306	211	0	122	89	1456	15617	1498	510
9	139	462	0	232	100	141	279	966	2451	836
10	1073	349	0	0	0	0	2012	1423	2420	6000

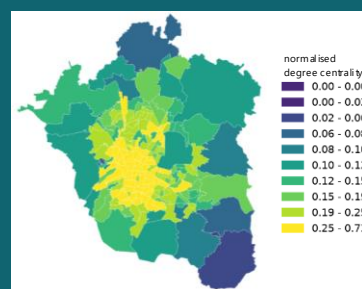
hourly change of population distribution



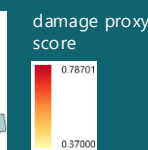
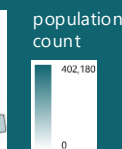
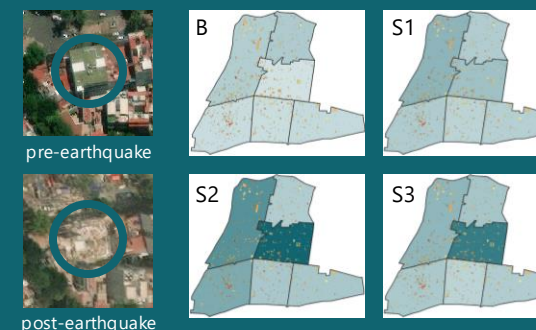
trip patterns



hourly graph degree centrality



emergency scenarios: damage assessment + population exposure



	"Tlatelolco" district		
	s1	s2	s3
Δpopulation vs baseline	-3.4%	-3.6%	+10.6%
total trips	51,851	21,707	21,238
degree centrality	0.34	0.23	0.22

damage data from the 19 September 2017 earthquake
B – Baseline (census data)
S1 – 1985 earthquake conditions: weekday, 7:00-7:59
S2 – 2017 earthquake conditions: weekday, 13:00-13:59
S3 – Tlatelolco maxpop scenario: weekend, 12:00-12:59

Dynamic population and mobility modelling reveals how exposure conditions in Mexico City Metropolitan Area shift by the hour, providing emergency managers and USAR teams with more operationally useful intelligence than static census data.